1. In a gaseous RbF molecule, the bond length is 2.274×10^{-10} m. Using data from Appendix F and making the oversimplified assumption as on the shape of the potential curve from $Rb^+ + F^-$ to an internuclear separation of 2.274×10^{-10} m, calculate the energy in kJ mol^{-1} required to dissociate RbF to neutral atoms.

2. Estimate the percent ionic character of the bond in each of the following species. All the species are unstable or reactive under ordinary laboratory conditions, but they can be observed in interstellar space.

	Bond Length (Å)	Dipole Moment (D)
OH	0.980	1.66
CH	1.131	1.46
CN	1.175	1.45
C_2	1.246	0

3. Draw Lewis electron dot diagrams for the following species, indicating formal charges and resonance diagrams where applicable.

   (a) HNC (central N atom)

   (b) SCN^- (thiocyanate ion)

   (c) H_2CNN (the first N atom is bonded to the carbon and the second N)

4. For each of the following molecules or molecular ions, give the steric number and sketch and name the approximate molecular geometry. In each case, the central atom is listed first and the other atoms are all bonded directly to it.

   (a) PF_3 (b) SO_2Cl_2 (c) PF_6^- (d) ClO_2^- (e) GeH_4

5. Ozone (O_3) has a nonzero dipole moment. In the molecule of O_3, one of the oxygen atoms is directly bonded to the other two, which are not bonded to each other.

   (a) Based on this information, state which of the following structures are possible for the ozone molecule: symmetric linear, nonsymmetric linear (for example, different O-O bond lengths), and bent. (Note: Even an O-O bond can have a bond dipole if the two oxygen atoms are bonded to different atoms or if only one of the oxygen atoms is bonded to a third atom.)

(b) Use the VSEPR theory to predict which of the structures of part (a) is observed.